INCH-POUND
MIL-PRF-3098/129D
7 August 2009
SUPERSEDING
MIL-PRF-3098/129C
17 May 2004

PERFORMANCE SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR148/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-3098.

Pertinent characteristics: 10 MHz to 61 MHz; third mechanical overtone; noncontrolled; series resonance.

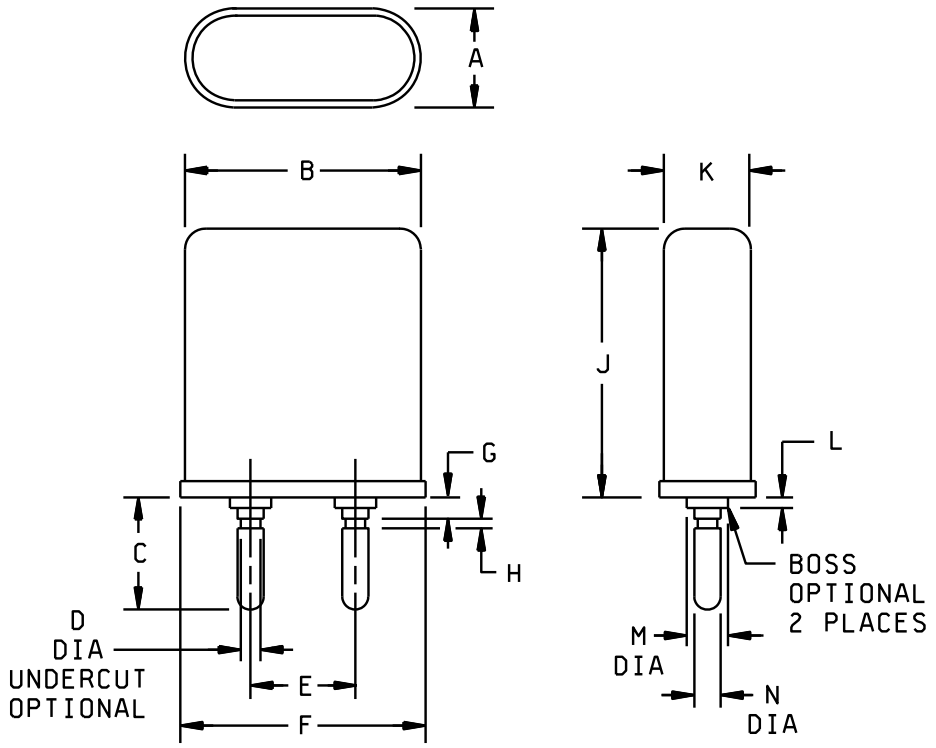

FIGURE 1. Crystal unit - CR148/U.

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Ltr	Inches		mm	
	Min	Max	Min	Max
A	---	.352	---	8.94
B	---	.725	---	18.42
C	.223	.248	5.66	6.30
D	.030	.037	0.76	0.94
E	.460	.476	11.68	12.09
F	---	.757	---	19.23
G	.030	.040	0.76	1.02
H	.015	.025	0.38	0.64
J	---	.775	---	19.68
K	---	.317	---	8.05
L	.015	.025	0.38	0.64
M	.075	.141	1.90	3.58
N	.048	.052	1.22	1.32

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking to be in accordance with MIL-PRF-3098.

FIGURE 1. Crystal unit - CR148/U - Continued.

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REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 10.0 MHz to 61.0 MHz, inclusive.

Capacitance, shunt: 7.0 pF, maximum.

Mode of oscillation: Third mechanical overtone.

Rated drive level: 1.0 mW, maximum.

Calibration values:

Equivalent resistance: 51 ohms.

Resistor voltage drop: 0.32 volt.

Resonance: Series.

Operating temperature range (noncontrolled): -55°C to +105°C, inclusive.

Frequency tolerance: ±0.0005 percent.

Equivalent resistance: 40 ohms, maximum.

Unwanted modes: Minimum unwanted-to-main mode resistance ratio of 2:1.

Shock:

Frequency change permitted: ±0.0005 percent.

Equivalent resistance change permitted: ±15 percent or 2 ohms, whichever is greater.

Vibration: Method 201, MIL-STD-202.

Frequency change permitted: ±0.0005 percent.

Equivalent resistance change permitted: ±15 percent or 2 ohms, whichever is greater.

Temperature cycling:

Frequency change permitted: ±0.0005 percent.

Equivalent resistance change permitted: ±15 percent or 2 ohms, whichever is greater.

Aging:

Frequency change permitted: ±0.0005 percent.

Reference documents. In addition to MIL-PRF-3098, this document references the following:

MIL-STD-202

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(Project 5955-2009-039)

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